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Dahod**

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Web Programming (3160713)

MINI – PROJECT REPORT

ON

**Online Blog Management System
(BlogSphere)**

Submitted By:

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**Under Guidance of
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ACKNOWLEDGEMENT

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PROJECT ABSTARCT

The primary objective of **BlogSphere** is to develop a comprehensive, student-friendly online blogging management system using procedural PHP and MySQL. In today's digital era, sharing information through blogs has become a fundamental part of communication. This project aims to simplify this process by providing a platform that handles all core aspects of blogging—from user registration and secure authentication to dynamic content creation and category-based organization.

Detailed Objectives:

- **Secure Authentication:** To implement a robust user authentication system that uses internal PHP security features like `password_hash()` and `password_verify()` to ensure user data privacy.
- **Dynamic Content Management:** To create a seamless CRUD (Create, Read, Update, Delete) experience that allows even non-technical users to manage their own blog posts without writing a single line of code.
- **Image Handling:** To demonstrate effective file handling by merging web-based image links with local file uploads. This involves handling temporary file storage, unique naming conventions, and persistent storage in the server's filesystem.
- **User Engagement:** To design a UI that is not only functional but also visually stunning. By using Bootstrap 5 and modern CSS techniques like glassmorphism, the project aims to set a high standard for student-level web applications.
- **Scalability:** To build a database structure that can easily be expanded in the future to include social features like comments, likes, and user profiles.

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1. PROBLEM STATEMENT

Traditional blogging platforms can be overly complex for beginners or lack the specific premium aesthetic required for a modern "wow" factor. Many student projects struggle with integrating multiple components like secure file uploads and database-driven filtering into a single cohesive unit.

Core Issues Addressed:

1. **Security Vulnerability:** Many basic blog systems store passwords in plain text, making them easy targets. BlogSphere addresses this by using Bcrypt hashing.
2. **UI Monotony:** Most administrative panels are dull and hard to navigate. BlogSphere uses a dark-neutral theme with vibrant accents and animations to enhance the experience.
3. **Complex Workflows:** Existing systems often require many steps to publish an image. BlogSphere simplifies this with a dual-input system (URL or File Upload).
4. **Data Redundancy:** Without proper categorization, blogs become disorganized. BlogSphere implements a strict relational database model to ensure every post is correctly classified.

2. TECHNOLOGIES USED

The selection of technologies for **BlogSphere** was driven by the need for performance, ease of use, and modern design capabilities.

2.1 Frontend (The User Interface)

- **HTML5:** Used for semantic structuring of web pages, ensuring accessibility and SEO friendliness.
- **CSS3 (Advanced):** Beyond basic styling, CSS3 was used for:
- **Glassmorphism:** Using backdrop-filter: blur() to create frosted-glass effects.
- **Micro-animations:** Smooth hover transitions and page fade-in effects.
- **Variables:** Using CSS variables for consistent HSL color palettes.
- **Bootstrap 5.3:** The newest version of Bootstrap was selected for its revamped grid system and lightweight components. It ensures the site is fully responsive on mobile, tablet, and desktop screens.
- **Google Fonts (Outfit):** A geometric sans-serif typeface was chosen to give the project a clean, premium, and professional look.

2.2 Backend (The Engine)

- **PHP 8.x:** As the world's most popular server-side language for web development, PHP was the natural choice. We used the procedural approach to match the academic curriculum while ensuring high-speed processing of form data and database queries.
- **Session Management:** PHP's native session handling is used to manage user states, track login status, and enforce role-based access control (Admin vs. User).

2.3 Database (The Brain)

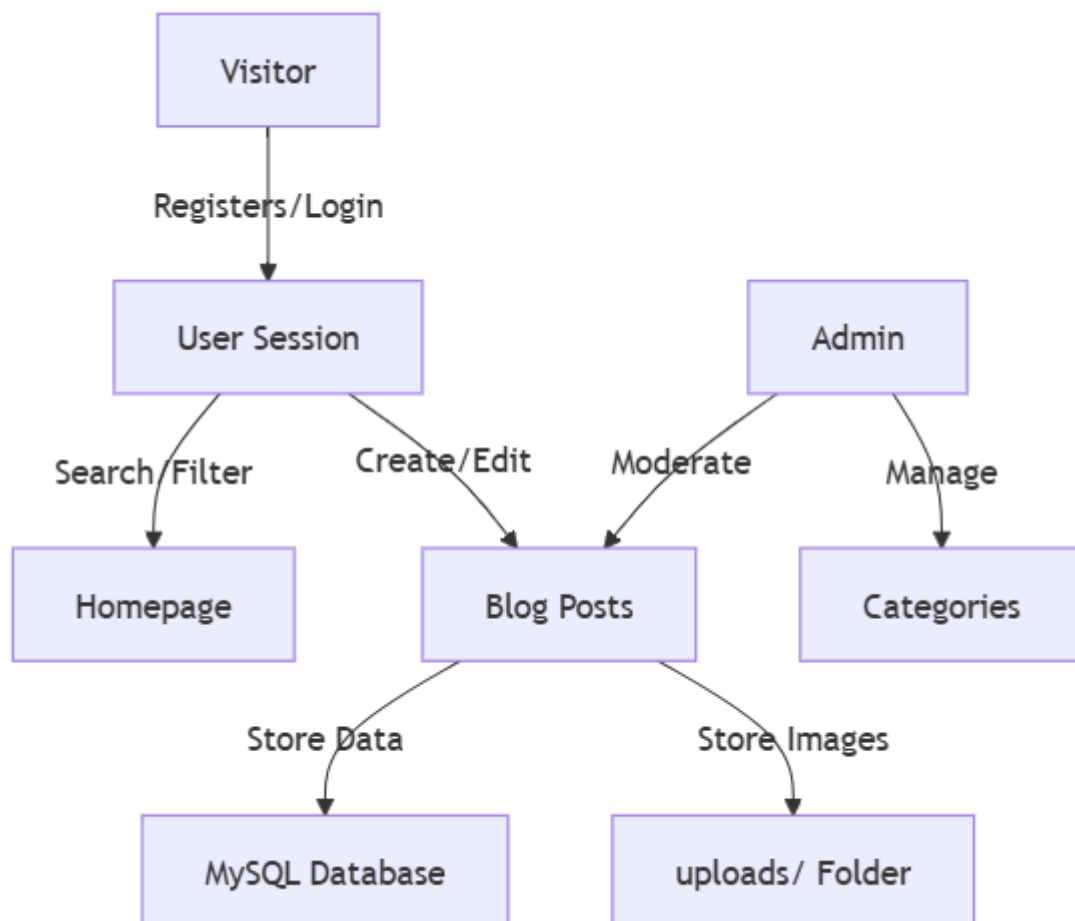
- **MySQL:** A relational database was chosen for its reliability and speed. It stores all data in three key tables: users, categories, and posts, linked through primary and foreign keys.

2.4 Development Tools

- **XAMPP:** For local server hosting (Apache & MySQL).
- **VS Code:** The primary Integrated Development Environment (IDE) used for code writing.
- **MySQL Workbench:** For visual database design and schema

3. SYSTEM ARCHITECTURE / WORKFLOW DIAGRAM

The system follows a typical **Client-Server Architecture**. When a user requests a page, the PHP engine processes the request, fetches data from the MySQL database, and renders the final HTML back to the browser.



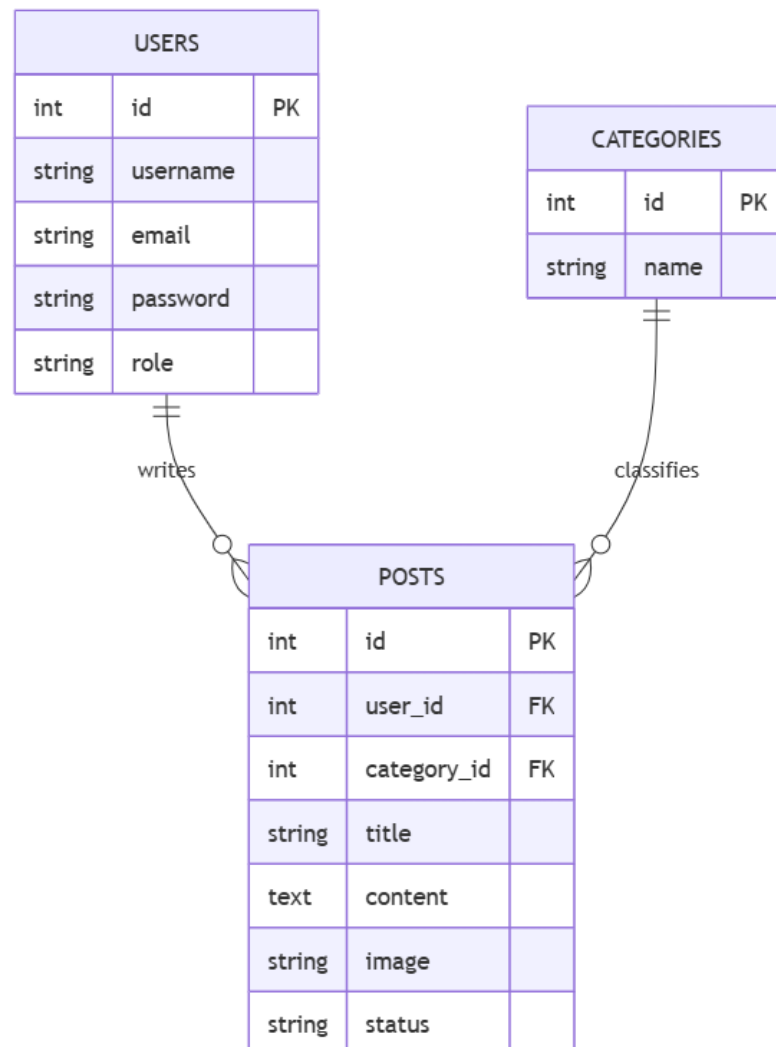
4. DATABASE DESIGN (TABLES)

The database blog_sphere_db is designed to be lean and efficient. All relationships are 1-to-Many.

4.1 Table Structures

1. **Users Table:** Stores id, username, email, password (hashed), and role (admin/user).
2. **Categories Table:** Stores id and name. Used for populating dropdowns and filtering results.
3. **Posts Table:** Stores id, user_id (FK), category_id (FK), title, content, image path/URL, and created_at.

4.2 ER Diagram



5. IMPLEMENTATION DETAILS

The implementation of **BlogSphere** is divided into five key phases:

5.1 Database Connectivity & Setup

The foundation of the project is the `includes/db.php` file. It establishes a persistent connection to the MySQL server and initializes the PHP session. Error handling is built-in to notify the developer if the database credentials (hostname, username, password) are incorrect.

5.2 User Authentication System

Authentication is handled via `login.php` and `register.php`.

- **Registration:** Before saving a user, the system checks if the email is already in use. It then uses `password_hash()` to encrypt the password.
- **Login:** The system fetches the hashed password from the database and compares it using `password_verify()`. If they match, `$_SESSION['user_id']` is set.

5.3 Dynamic Blog Management (CRUD)

This is the heart of the application.

- **Creation:** `create_post.php` handles a multipart form. If a file is uploaded, it is moved to the `uploads/` folder using `move_uploaded_file()`. Otherwise, the image URL provided by the user is used.
- **Display:** The homepage uses an SQL JOIN to fetch post data along with the author's name and category title in a single query.
- **Editing:** `edit_post.php` pre-fills the form with existing data, allowing users to update their content without re-typing everything.

5.4 Administrative Controls

Administrators have access to a dedicated dashboard. Their primary task is **Category Management**. They can add new categories (e.g., "AI & ML", "Lifestyle") which then automatically appear in the user's category dropdown during post creation.

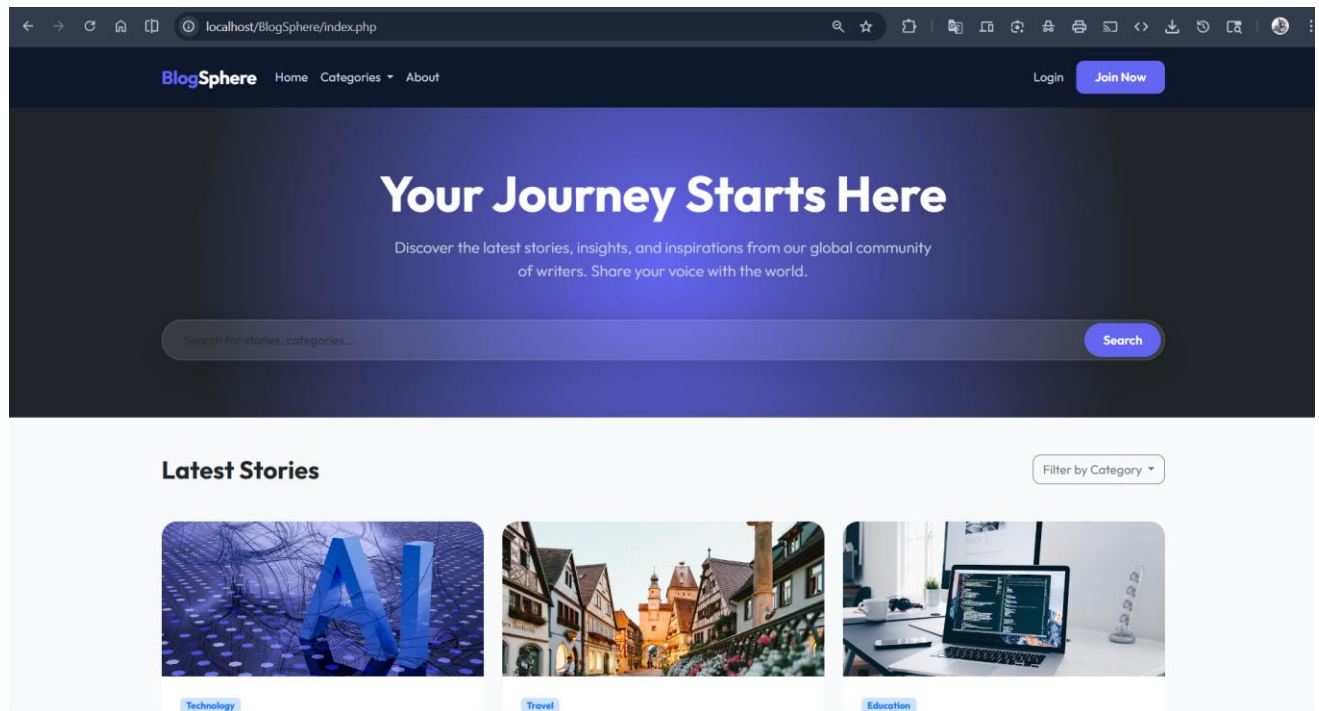
5.5 UI/UX Design System

We implemented a custom CSS file tied into `includes/header.php`. This includes:

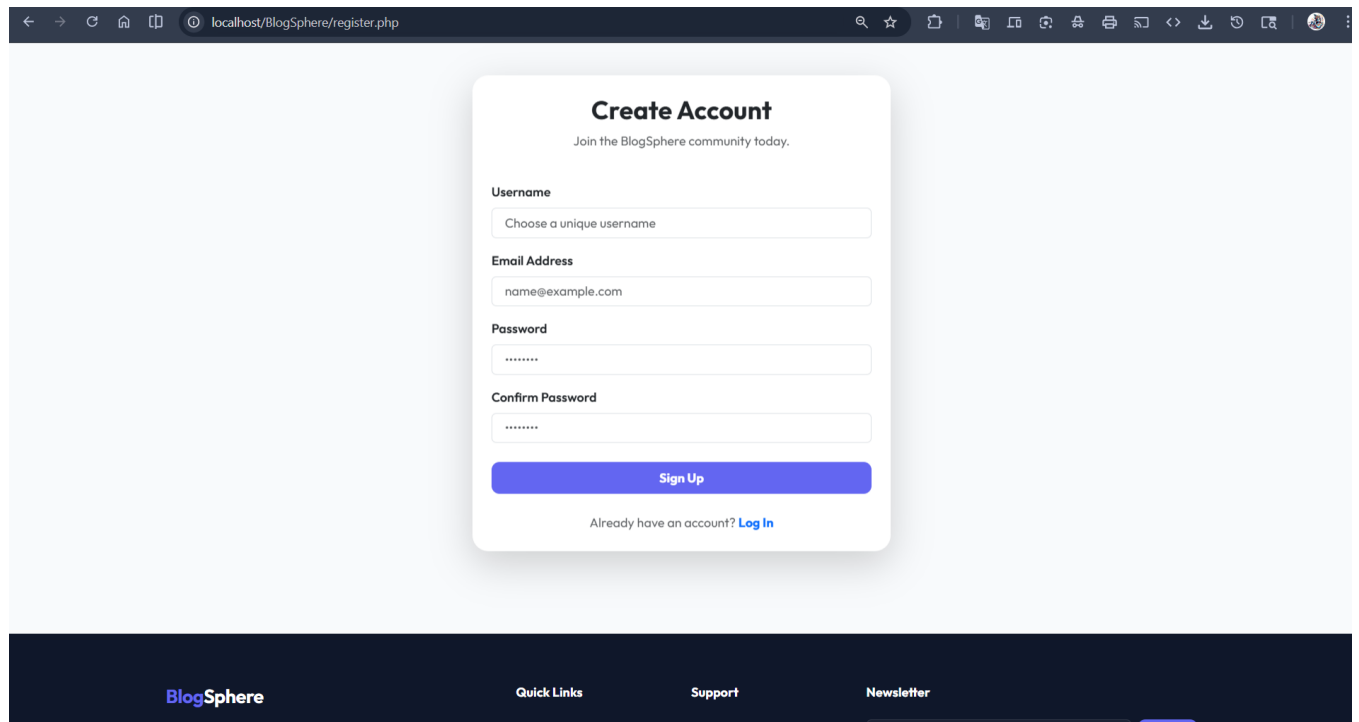
- **HSL Colors:** For precise color control and vibrant look.
- **Hover Transitions:** Blog cards slightly scale up and gain a shadow when hovered.
- **Navigation:** A responsive navbar that shows "Login/Signup" for visitors and "My Stories/Logout/Dashboard" for logged-in users.

6. SCREENSHOTS OF THE PROJECT

1. **Homepage:** Showcase the hero section, the glassmorphism search bar, and the blog card grid.



2. **Registration Page:** Show the clean design and validation messages.



3. Login Page: Display the minimalist login form.

localhost/BlogSphere/login.php

Welcome Back

Sign in to continue your writing journey.

Username or Email

Password [Forgot?](#)

[Log In](#)

Don't have an account? [Sign Up](#)

BlogSphere
A platform for writers to share their stories and for readers to explore new perspectives. Built for Web Programming Course.

Quick Links
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Newsletter

[Sub](#)

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4. User Dashboard: Show the table view of the user's personal posts with "Edit" and "Delete" actions.

localhost/BlogSphere/dashboard.php

BlogSphere [Home](#) [Categories](#) [About](#) [raj](#)

Hello, raj!

Welcome back to your creator dashboard.

[+ Create New Post](#)

TOTAL STORIES
2

PUBLISHED
2

DRAFTS
0

Recent Activities [Manage All Posts](#)

Title	Category	Status	Date	Actions
Top 5 Hidden Gems in Europe	Travel	Published	Apr 22, 2026	Edit Delete
The Art of Slow Living	Lifestyle	Published	Apr 22, 2026	Edit Delete

BlogSphere [Home](#) [Categories](#) [About](#) [raj](#)

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5. Create Post Form: Highlight the dual image input (URL and File Upload).

The screenshot shows a web browser at `localhost/BlogSphere/create_post.php`. The page has a dark blue header with the 'BlogSphere' logo and navigation links: Home, Categories, and About. A user profile 'raj' is in the top right. The main content is a white modal titled 'Create New Post' with the subtitle 'Share your thoughts with the community.' The form includes: a 'Post Title' input field with placeholder 'Enter a catchy title'; a 'Category' dropdown menu with 'Select Category'; a 'Featured Image URL' input field with placeholder 'https://example.com/image.jpg'; an 'Or Upload from PC' section with 'Choose File' and 'No file chosen' buttons; a 'Content' text area with placeholder 'Write your story here...'; and a 'Status' section with radio buttons for 'Save as Draft' (selected) and 'Publish Now'. At the bottom are 'Save Post' and 'Cancel' buttons.

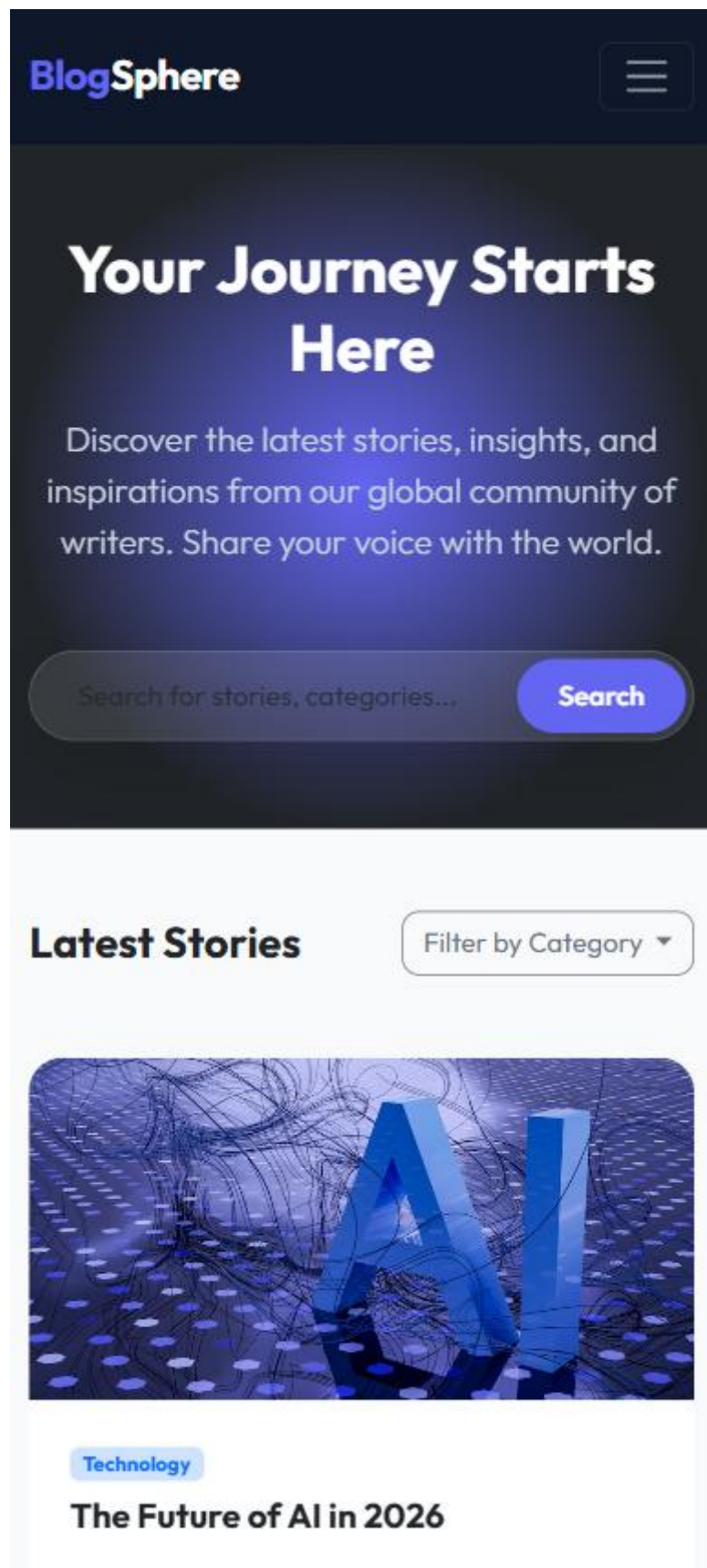
6. Admin Category Page: Show how an admin can manage the blog's topics.

The screenshot shows a web browser at `localhost/BlogSphere/admin/categories.php`. The page has a blue header with 'Admin Panel' and navigation links: Dashboard, Categories, and Website. The main content area has two sections: 'Add New Category' and 'Existing Categories'. The 'Add New Category' section has a 'Category Name' input field with placeholder 'e.g. Fashion, Sports' and an 'Add Category' button. The 'Existing Categories' section contains a table with the following data:

ID	Name	Created At	Action
6	Health	2026-04-22 11:30:25	
5	Food	2026-04-22 11:30:25	
4	Education	2026-04-22 11:24:00	
3	Travel	2026-04-22 11:24:00	
2	Lifestyle	2026-04-22 11:24:00	
1	Technology	2026-04-22 11:24:00	

The footer is dark blue and contains the 'BlogSphere' logo, a description: 'A platform for writers to share their stories and for readers to explore new perspectives. Built for Web Programming Course.', 'Quick Links' (Home, Login, Register), 'Support' (About Us, Contact, Terms), and a 'Newsletter' section with an input field and a 'Sub' button. A copyright notice '© 2026 BlogSphere. Developed by Raj Patel.' is at the bottom center.

7. **Mobile View:** Demonstrate the site's responsiveness on a smartphone screen.



7. CONCLUSION

The development of **BlogSphere** was a significant learning experience that combined the core principles of the Web Programming curriculum with modern industry trends.

Critical Learnings:

- **Full-Stack Integration:** Understanding how front-end HTML/CSS interacts with back-end PHP and a MySQL database.
- **Security Awareness:** Learning how to defend against common vulnerabilities like SQL Injection and plain-text password storage.
- **User-Centric Design:** Realizing that a project is only as good as its user experience, leading to the inclusion of animations and glassmorphism. In summary, the project successfully delivered a robust, secure, and visually appealing blog management system that is ready for real-world academic submission.

8. REFERENCES

1. **PHP Manual:** Official documentation for procedural MySQLi and session management (php.net).
2. **Bootstrap:** Official documentation for grid systems and components (getbootstrap.com).
3. **W3Schools:** Tutorials on PHP loops, arrays, and MySQL queries.
4. **YouTube/StackOverflow:** For debugging common PHP image upload errors and CSS animation techniques.